OPENCV

OpenCV

OpenCV (Open Source Computer Vision Library) is an open-source library used for computer vision and image processing tasks.
It supports real-time applications and works with Python, C++, and Java.

Computer Vision

Computer Vision is a branch of Artificial Intelligence that enables computers to understand and interpret images and videos.

Features of OpenCV:

- Real-time image and video processing
- Face detection and recognition
- Object detection and tracking
- Image filtering and transformation
- Edge detection
- Motion detection
- Deep learning support (DNN module)

Applications of OpenCV:

- Face Recognition
- Object Detection
- Motion Tracking
- Medical Imaging
- OCR
- Augmented Reality

Important OpenCV Functions (Python)

import cv2

- cv2.imread() → Read image
- cv2.imshow() → Display image
- cv2.imwrite() → Save image
- cv2.VideoCapture() → Capture webcam video
- cv2.cvtColor() → Convert color format
- cv2.CascadeClassifier() → Face detection

Example

```
import cv2
img = cv2.imread("image.jpg")
cv2.imshow("Image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```